Wades Cave

Survey Report

Greenbrier County, West Virginia

DRAFT · version 0.1 (draft) · 2026-11-02

Draft for review by Mystik Miller and Bill Balfour. Not for distribution. Survey data belong to the landowner(s).

Compiled by Aaron Bird, data and cartography, Wades Cave Survey Project.
Generated from the survey data; see the change log at the back.

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Coordinates in this draft are given relative to the entrance station GG011 (feet east, north and up). Georeferenced coordinates are kept in the survey and GIS data files.

1. Summary

Wades Cave, closed to trips since the 1980s and not surveyed since the 1970s, was reopened to a Greenbrier Grotto survey project in 2026. The first trip (2026-10-31) surveyed **218 ft** in 10 legs from GG001 out to the entrance (GG011), with two instruments and frontsights and backsights on every leg.

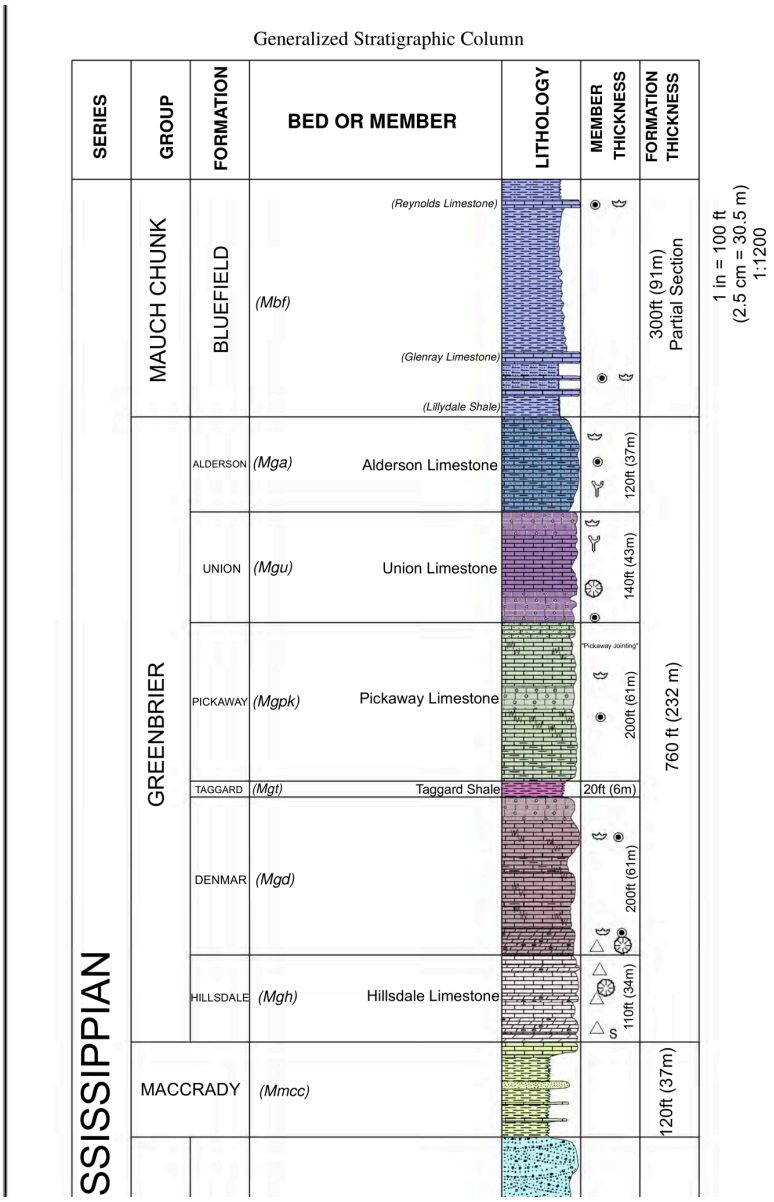
Item	Value
Survey to date	218.0 ft (1 trip, 10 legs, 11 stations)
Vertical range of stations	-29.0 to +0.0 ft relative to GG011
Entrance tie	GG011 at the entrance GPS fix, 4 ft below ground
1960s/70s lineplot	about 16,400 ft of line (digitized from the old map)
Open leads	to be compiled (section 8)

PLACEHOLDER · Summary text

- Two or three paragraphs: project purpose, partners, what the survey has found so far.
- Running totals table per trip (footage, stations, leads) once more trips are in.

2. Location and geologic setting

Wades Cave lies near Lewisburg in Greenbrier County, east of US 219. It is one of the contact caves, developed in the lowest units of the Greenbrier Group (Hillsdale Limestone and Denmar Formation) at their contact with the Maccrady Shale. The bedrock geology is from WVGES OF-2103, *Bedrock Geologic Map of the Lewisburg 7.5' Quadrangle* (Tudek et al., 2022). Per OF-2103, the Hillsdale and Denmar host most of the karst, fed by water draining off the Maccrady, and the area drains to Davis Spring.



Generalized stratigraphic column, WVGES OF-2103 (Lewisburg quadrangle). Wades Cave is in the Hillsdale and Denmar, just above the Maccrady.

3. History of exploration and survey

PLACEHOLDER · History

- 1960s and 1970s surveys: per Bill Balfour, the old lineplot combines survey from both decades; which parts came from which survey is not known.
- Source of the old lineplot: West Virginia Association for Cave Studies sheet The Contact Caverns of the Great Savannah.
- Closure in the 1980s, landowner relations, reopening and the 2026 Greenbrier Grotto project.
- Names of earlier surveyors and explorers to credit (Bill Balfour to advise).

4. Maps

4.1 Overview

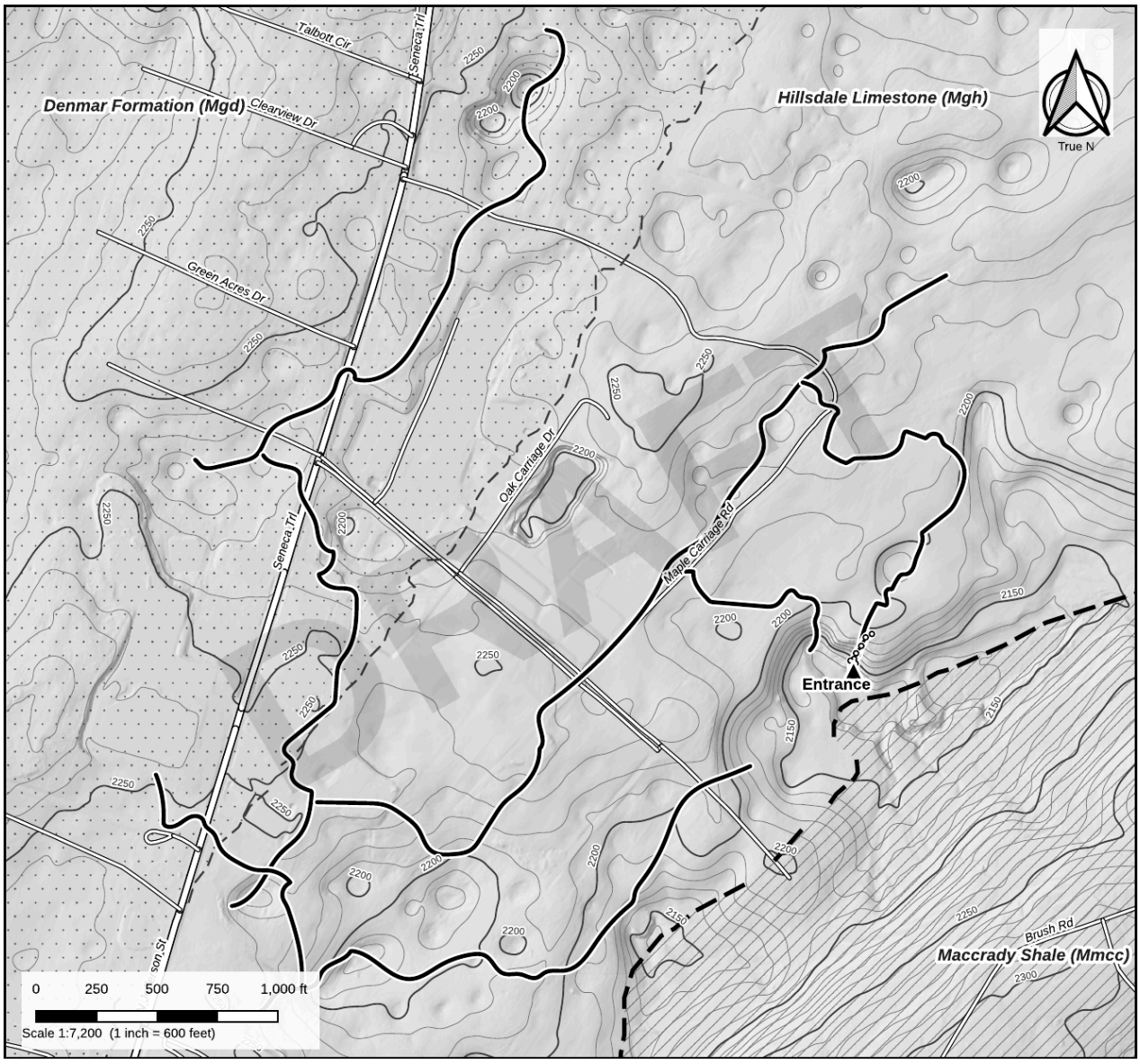
The full overview map (lidar topography, bedrock geology, 1960s/70s lineplot and the 2026 survey) follows on the next page. An 11x17 edition is planned.

WADES CAVE

1960s lineplot on lidar topography and bedrock geology · Greenbrier Co., WV

DRAFT VERSION

2026-11-02 · entrance GPS 2026-10-31



- Wades Cave lineplot (1960s)
- New survey GG, 2026-10-31
- Entrance (GPS)

Geology (WVGES OF-2103)

- Limestone / shale contact (Hillsdale / Maccrady)
- Formation contact

Bedrock units

- Maccrady Shale (Mmcc) - non-carbonate
- Hillsdale Limestone (Mgh) - no pattern
- Denmar Formation (Mgd) - limestone

Contours

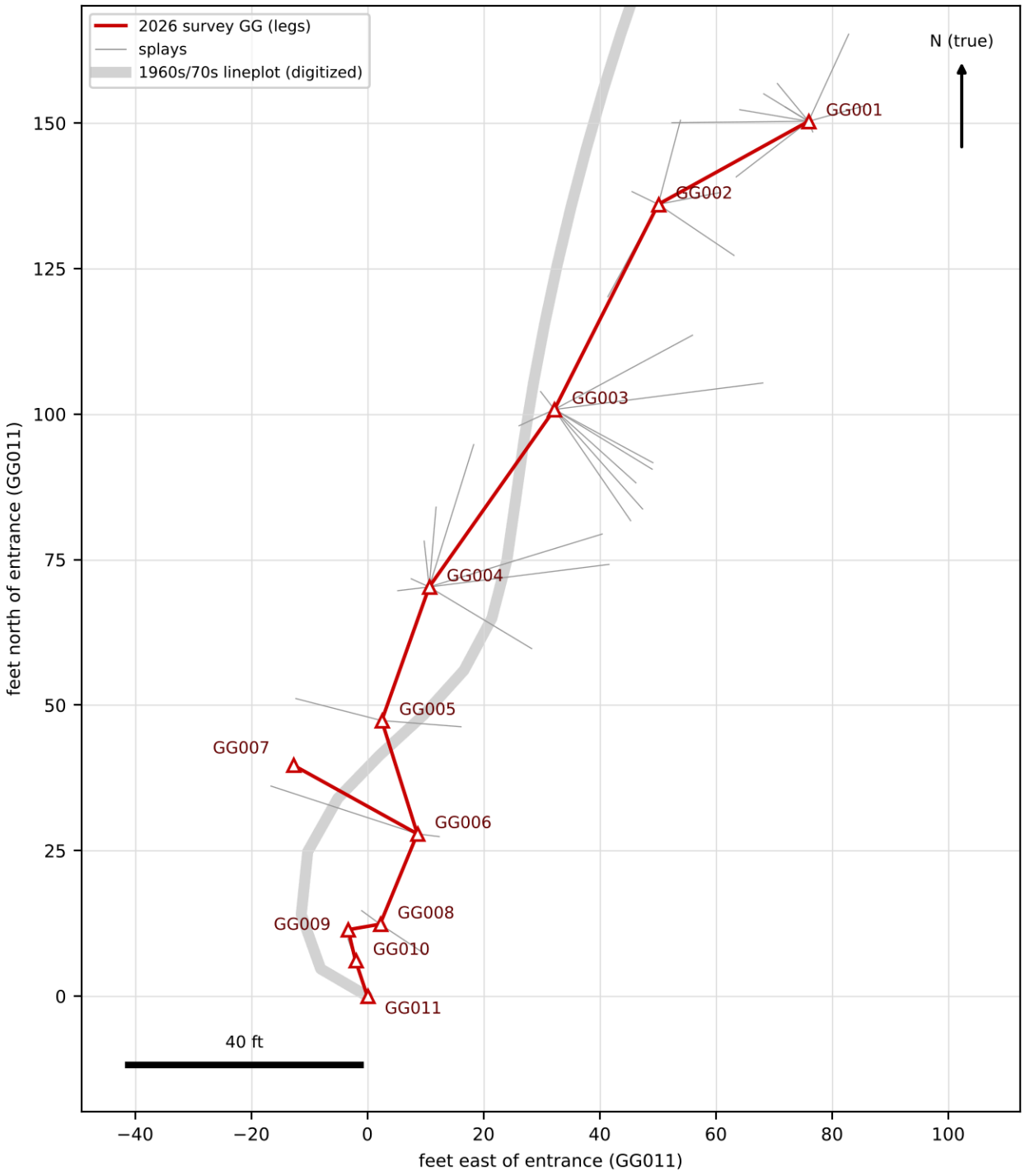
- Index contour (50 ft)
- Contour (10 ft)

Roads

- Local road
- Highway (US 219)

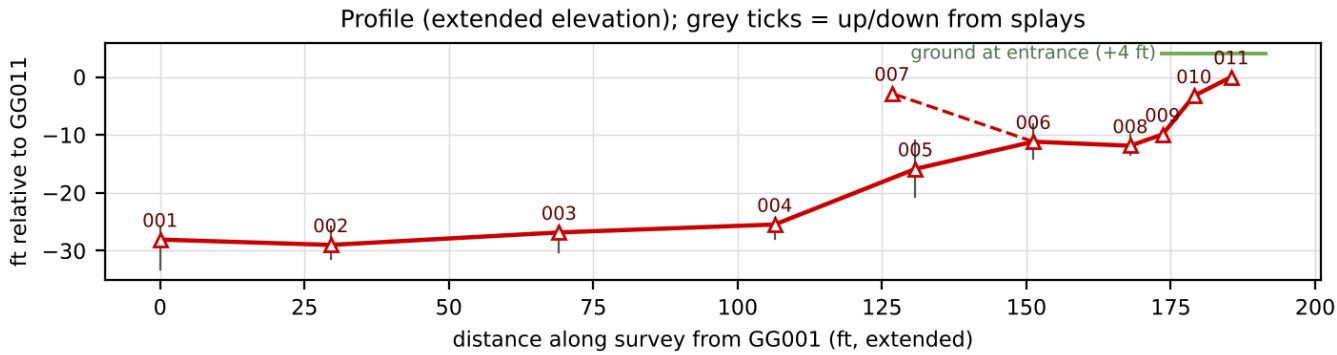
New survey: GG001–GG011, 218 ft, 2026-10-31 (BRIC5-0507 and BRIC4-0038, frontsights and backsights averaged, declination −8.6°), tied to the entrance GPS; it follows the 1960s entrance passage within about 4–12 ft. **Original survey:** 1960s lineplot as compiled on the West Virginia Association for Cave Studies sheet *The Contact Caverns of the Great Savannah*; survey team and date to be credited. Scaled from its 1000 ft bar; page-up taken as north. **Positioning:** tied to the entrance by GPS on 2026-10-31 (mean of Garmin GPSMAP 64s and iPhone fixes, 40 ft apart); rotation not yet checked against a second point, so passage positions far from the entrance are approximate until a second tie point is surveyed.
Topography: USGS 3DEP 1 m lidar (2016) hillshade; 10 ft contours, 50 ft index labeled in feet. Roads: USGS National Transportation Dataset. NAD83 / UTM 17N.
Geology: WVGES OF-2103, *Bedrock Geologic Map of the Lewisburg 7.5′ Quadrangle* (Tudek et al., 2022, 1:24,000), traced from the published map; contacts approx. ±15–30 m. Per OF-2103, the Hillsdale and Denmar host most karst, fed by runoff off the Maccrady; the area drains to Davis Spring.
Compiled by Aaron Bird, Wades Cave Survey Project, Greenbrier Grotto, 2026-11-02. DRAFT – not for distribution; survey data belong to the landowner(s).

4.2 Entrance area plan



2026 survey (legs and splays) with the old lineplot behind it. Coordinates in feet from GG011; see the scale bar.

4.3 Profile

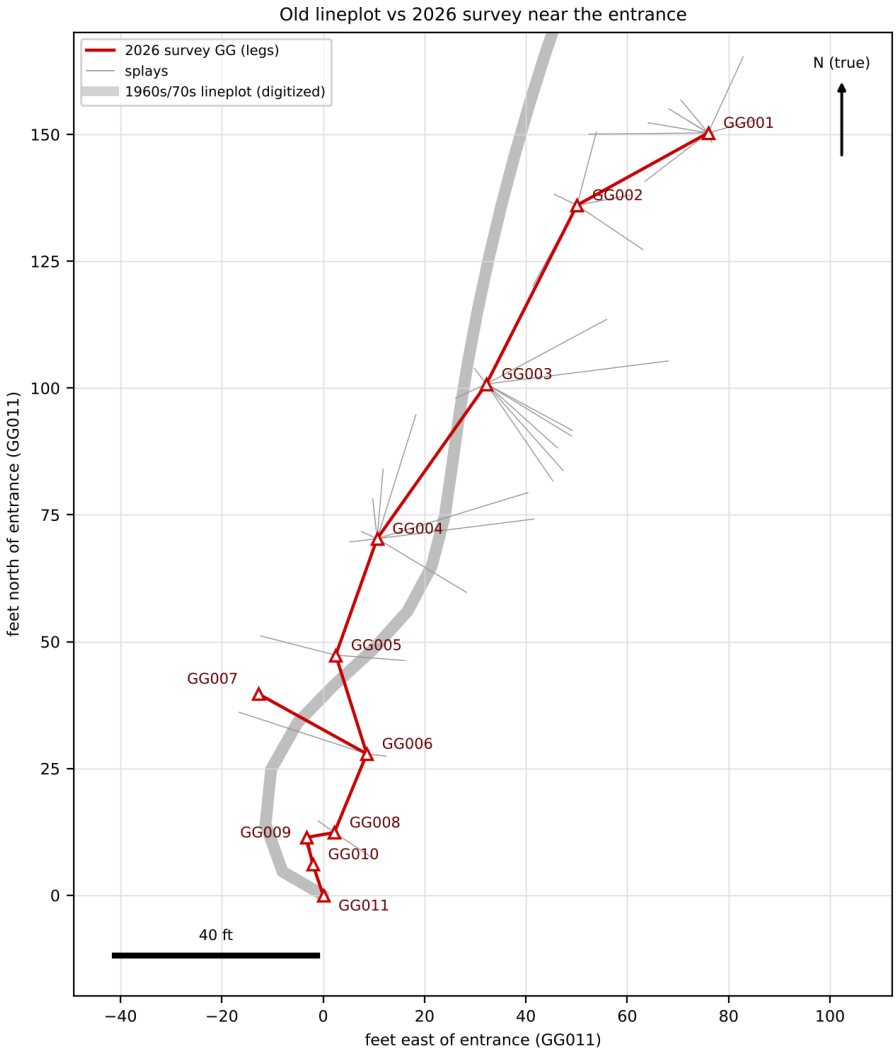


Extended elevation along GG001 to GG011; GG007 (dashed) is the side station off GG006.

PLACEHOLDER · Profile from sketches

- Replace the station profile with walls drawn from the profile sketches when they come in.

4.4 Old vs new comparison



The old lineplot is placed by the entrance GPS fix, scaled from its 1000 ft bar, with page-up taken as true north. From the entrance to GG005 the new survey lies within about 4 to 12 ft of the old line. A best-fit rotation of the old lineplot about the entrance is −3°, improving the RMS fit only from 9.9 to 9.2 ft, so it has not been rotated. A radiolocation (section 9) will allow a proper check.

Station	GG001	GG002	GG003	GG004	GG005	GG006	GG008	GG009	GG010	GG011	GG007
Distance to old lineplot (ft)	35.3	14.4	4.3	11.8	4.1	14.1	11.8	6.7	4.3	0.0	9.6

5. Survey methods and quality

Instruments

Instrument	Use	Recording
BRIC5-0507	all legs (frontsight triples), backsights, splays	TopoDroid 6.4.12, station names
BRIC4-0038	second frontsights and backsights	internal memory, matched to legs by value

Reduction

- Each leg is the average of all its observations with equal weight: the BRIC5 frontsight triple (as one mean), each BRIC5 backsight, and each BRIC4 frontsight and backsight. Backsights are reversed before averaging.
- GG001–GG002 and GG006–GG008: the triple shot farthest from the other two was dropped.
- Declination −8.62° (World Magnetic Model 2025, entrance, 2026-10-31). No instrument corrections: no course test was recorded.
- Left and right were taken facing outward, toward the entrance. LRUD values are the horizontal (L/R) or vertical (U/D) part of the labeled splays.
- Splays that could not be clearly associated with a station were left out (Appendix A notes).
- The entrance station GG011 is fixed at the GPS position: the mean of a Garmin GPSMAP 64s and an iPhone GPS Kit fix, about 40 ft apart. It is 4 ft below the ground surface.

Quality

- BRIC4-0038 reads about 1.3° lower in azimuth than BRIC5-0507 on the same legs; averaging splits the difference. Course tests of both instruments are recommended before every trip.
- Largest leg disagreements: GG008–GG009 3.5° azimuth (a 5.9 ft leg); GG003–GG004 2.3° azimuth and 0.47 ft distance.
- Shot 48 (BRIC5 file) was a hand-typed copy of BRIC4-0038 shot 7 with the clino typed as +0.20; the instrument recorded −1.97, which was used.
- The reduction was cross-checked by running the Survex file through Survex (cavern): station positions agree to the centimeter.

6. Trip 1: 2026-10-31 (GG survey)

6.1 Trip report and team

PLACEHOLDER · Trip report and team

- Team members and roles (book, instruments, point, sketch); TopoDroid team field was empty.
- Time in and out, objective, what was done, conditions, observations, open leads, hazards.
- Instrument course tests (none recorded for this trip).

6.2 Averaged legs

Distances in feet (meters alongside); azimuths true and magnetic; spreads are the range of the averaged observations.

From	To	Dist ft	Dist m	Azi true	Azi mag	Incl	Obs	Spread ft	Spread az°	Spread inc°
GG001	GG002	29.56	9.009	241.1	249.8	-1.8	5	0.05	1.1	0.3
GG002	GG003	39.58	12.065	206.9	215.5	+3.1	4	0.21	1.6	0.3
GG003	GG004	37.37	11.390	215.2	223.9	+2.1	4	0.47	2.3	0.3
GG004	GG005	26.19	7.983	199.6	208.2	+21.5	4	0.23	2.0	0.5
GG005	GG006	20.94	6.383	162.5	171.2	+13.0	4	0.03	1.4	0.4
GG006	GG007	25.84	7.876	298.9	307.6	+18.8	3	0.10	0.6	0.3
GG006	GG008	16.80	5.121	202.4	211.1	-2.4	5	0.28	1.8	0.6
GG008	GG009	5.92	1.806	260.2	268.8	+19.2	4	0.13	3.5	0.9
GG009	GG010	8.59	2.619	166.4	175.1	+50.7	3	0.07	0.7	1.3
GG010	GG011	7.23	2.204	161.5	170.2	+26.9	2	0.04	0.1	0.1

6.3 Stations and LRUD

Positions in feet relative to the entrance station GG011. LRUD in feet, left and right facing outward toward the entrance; "—" = not measured. GG008 left is 2 ft per the shot comment (the splay itself measured 9.3 ft).

Station	East ft	North ft	Up ft	Left	Right	Up	Down
GG001	+75.9	+150.3	-28.1	2.0	8.4	2.1	5.1
GG002	+50.1	+136.0	-29.0	15.7	5.0	3.1	2.4
GG003	+32.2	+100.8	-26.8	19.7	3.9	1.0	3.5
GG004	+10.6	+70.3	-25.5	20.6	3.4	1.2	2.5
GG005	+2.5	+47.3	-15.9	13.7	15.3	5.0	4.8
GG006	+8.6	+27.9	-11.2	3.8	26.5	3.1	2.9
GG007	-12.8	+39.7	-2.8	—	—	—	—
GG008	+2.2	+12.4	-11.9	2.0	3.9	1.9	1.5
GG009	-3.3	+11.4	-9.9	—	—	—	—
GG010	-2.0	+6.1	-3.3	—	—	—	—
GG011	+0.0	+0.0	+0.0	—	—	—	—

6.4 Splays

Station	Shot	Dist ft	Azi true	Incl	Label
GG001	4	1.95	158.3	+0.2	left
GG001	5	8.42	320.4	-0.0	right
GG001	6	2.10	336.1	+89.0	up
GG001	7	5.13	279.1	-84.5	down
GG001	8	15.70	232.5	+0.5	
GG001	9	23.58	269.4	-4.3	

Station	Shot	Dist ft	Azi true	Incl	Label
GG001	10	12.04	279.6	-3.9	
GG001	11	4.67	301.5	-2.5	
GG001	12	9.06	301.5	-2.5	
GG001	13	9.88	74.6	-8.8	
GG001	14	16.56	24.8	-5.0	
GG002	21	15.85	123.9	-7.5	left
GG002	22	5.01	295.9	+2.6	right
GG002	23	3.16	191.5	+83.3	up
GG002	24	2.43	200.3	-87.6	down
GG002	25	10.79	79.3	+6.3	
GG002	26	14.95	14.8	-0.0	
GG002	27	18.15	208.7	+3.8	
GG003	29	36.32	82.8	-5.1	(splay; commented 'back')
GG003	30	27.11	61.8	-5.5	
GG003	31	19.42	118.3	-7.1	
GG003	32	18.99	131.9	-6.7	
GG003	33	23.02	138.4	-6.3	
GG003	34	23.31	145.5	-5.9	
GG003	35	6.72	245.2	-4.8	
GG003	36	18.73	26.4	-2.2	
GG003	37	19.82	121.4	-6.0	left
GG003	38	3.89	322.4	+1.1	right
GG003	39	0.99	144.1	+77.2	up
GG003	40	3.49	14.8	-82.3	down
GG004	49	20.69	121.0	-5.8	left
GG004	50	3.40	294.4	-1.7	right
GG004	51	1.24	81.6	+83.3	up
GG004	52	2.45	80.4	-88.0	down
GG004	53	31.30	73.0	-5.8	
GG004	54	31.33	82.9	-5.5	
GG004	55	25.73	17.4	-5.0	
GG004	56	13.91	4.9	-8.1	
GG004	57	8.03	353.7	-9.4	
GG004	58	5.97	263.2	+24.1	
GG005	63	13.71	94.4	+2.5	left
GG005	64	15.41	284.3	+7.9	right
GG005	65	4.98	83.2	+85.4	up
GG005	66	4.84	90.6	-85.5	down
GG006	70	3.81	97.1	-2.2	left
GG006	71	28.15	288.1	+19.7	right
GG006	72	3.10	186.8	+82.4	up
GG006	73	2.95	19.9	-83.3	down
GG008	85	9.35	123.5	+0.4	left is 2 feet
GG008	86	3.94	306.2	-2.2	right
GG008	87	1.96	15.4	+85.2	up
GG008	88	1.52	318.3	-87.5	down
GG010	99	1.61	339.9	-26.3	

Station	Shot	Dist ft	Azi true	Incl	Label
GG010	100	4.44	340.7	-27.2	

6.5 Sketches

PLACEHOLDER · Sketches

- Scanned plan, profile and cross-section sketches for GG001 to GG011.
- Entrance-area plan with walls drawn from the sketches.

7. Photo log

Every photo is tied to a location: a survey station or an **OLDMAP** photo point. Photo points are places visited but not yet surveyed; they are numbered on the old map and the photo file carries the number (OLDMAP001.JPEG). Their positions are approximate, taken from the marks on the old lineplot. When the survey reaches a photo point, the table records its nearest station and the point moves to surveyed coordinates. In the web version a camera icon at each point will open its photos and Live Photo clips.

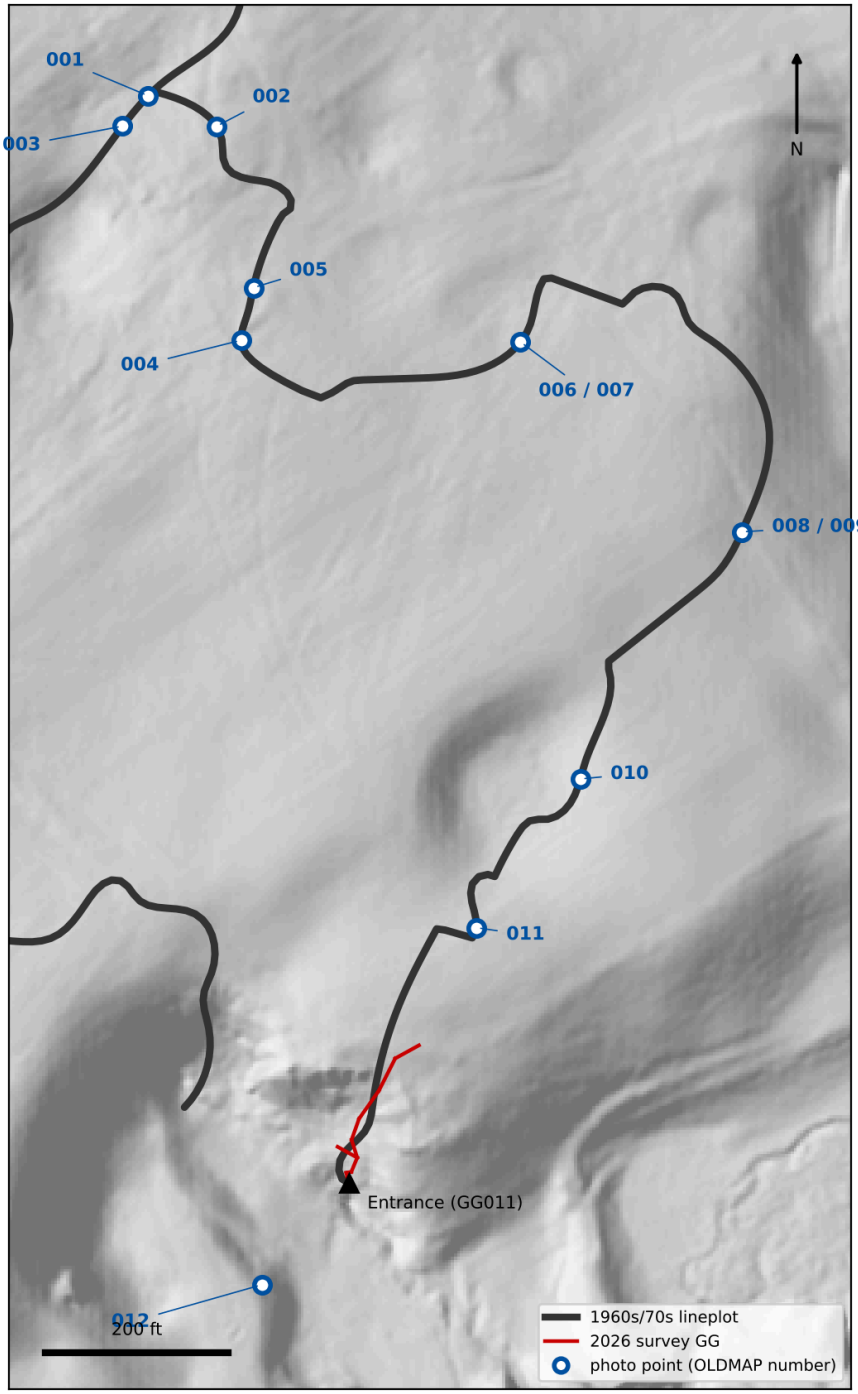


Photo points from trip 1 (2026-10-31), marked by A. Bird on the old map. The 2026 survey (red) so far reaches only the first ~170 ft inside the entrance.

Photo points, trip 1 (2026-10-31)

Times from the photos (EDT). Positions in feet east / north of the entrance station GG011. Live clips are the 2–3 s Live Photo videos taken with each photo.

Point	Time	Camera	Live clip	E / N ft	On	Note
OLDMAP001	16:06	iPh 15 Pro Max	—	-219 / +1186	old lineplot	
OLDMAP002	16:07	iPh 14 Pro	IMG_5031.MOV	-145 / +1152	old lineplot	
OLDMAP003	16:13	iPh 14 Pro	—	-247 / +1152	old lineplot	
OLDMAP004	15:52	iPh 14 Pro	—	-118 / +919	old lineplot	
OLDMAP005	15:53	iPh 14 Pro	IMG_5012.MOV	-105 / +976	old lineplot	
OLDMAP006	15:36	iPh 15 Pro Max	IMG_5925.MOV	+187 / +917	old lineplot	
OLDMAP007	15:45	iPh 15 Pro Max	IMG_5927.MOV	+187 / +917	old lineplot	same spot as OLDMAP006
OLDMAP008	15:08	iPh 15 Pro Max	IMG_5911.MOV	+428 / +710	old lineplot	
OLDMAP009	15:07	iPh 15 Pro Max	IMG_5907.MOV	+428 / +710	old lineplot	same spot as OLDMAP008
OLDMAP010	14:56	iPh 14 Pro	IMG_4999.MOV	+252 / +441	old lineplot	
OLDMAP011	14:56	iPh 15 Pro Max	IMG_5900.MOV	+138 / +279	old lineplot	
OLDMAP012	12:26	iPh 15 Pro Max	IMG_5878.MOV	-96 / -111	surface	surface, near the entrance; the Live Photo GPS (+/-19 m) puts it ~50 ft from the entrance fix, 180 ft from this mark

Photos

PLACEHOLDER · Captions

- Caption and photographer for each photo; what is shown (passage, feature, people).



OLDMAP001 · 16:06 · iPhone 15 Pro Max
Caption to come.



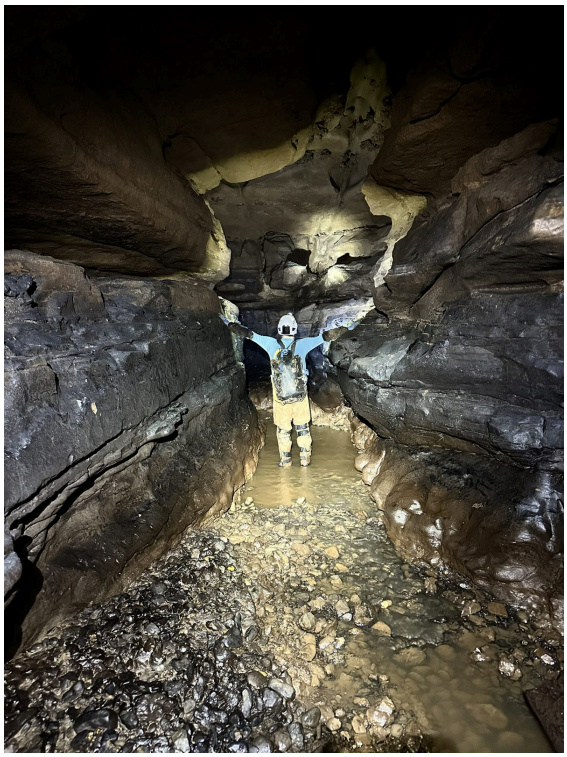
OLDMAP002 · 16:07 · iPhone 14 Pro · Live clip IMG_5031.MOV
Caption to come.



OLDMAP003 · 16:13 · iPhone 14 Pro
Caption to come.



OLDMAP004 · 15:52 · iPhone 14 Pro
Caption to come.



OLDMAP005 · 15:53 · iPhone 14 Pro · Live clip IMG_5012.MOV
Caption to come.



OLDMAP006 · 15:36 · iPhone 15 Pro Max · Live clip IMG_5925.MOV
Caption to come.



OLDMAP007 · 15:45 · iPhone 15 Pro Max · Live clip IMG_5927.MOV
Caption to come.



OLDMAP008 · 15:08 · iPhone 15 Pro Max · Live clip IMG_5911.MOV
Caption to come.



OLDMAP009 · 15:07 · iPhone 15 Pro Max · Live clip IMG_5907.MOV
Caption to come.



OLDMAP010 · 14:56 · iPhone 14 Pro · Live clip IMG_4999.MOV
Caption to come.



OLDMAP011 · 14:56 · iPhone 15 Pro Max · Live clip IMG_5900.MOV
Caption to come.



OLDMAP012 · 12:26 · iPhone 15 Pro Max · Live clip IMG_5878.MOV
Caption to come.

8. Leads

#	Station	Type (?, TT, TD, climb, dig)	Description	Size	Priority	Status
1						
2						
3						
4						
5						
6						
7						
8						

PLACEHOLDER · Leads

- Filled from the trip reports and sketches; GG007 purpose to be recorded.

9. Radiolocation and survey control

Control	Status
Entrance GPS (GG011)	2026-10-31: Garmin GPSMAP 64s and iPhone GPS Kit, mean used; fixes 40 ft apart
Radiolocation	Planned once the survey reaches the main passages
Old lineplot fit	Placed by entrance only; rotation to be checked after radiolocation

PLACEHOLDER · Radiolocation

- Ground-zero GPS, cave station, depth, closure, and the refit of the old lineplot.

Appendix A: raw shots, BRIC5-0507

All shots as recorded in TopoDroid (meters, magnetic azimuth), with feet alongside and how each was used.

Shot	Role	From	To	Dist (m)	Dist (ft)	Azi (mag)	Clino	Used	Note
1	FS	GG001	GG002	8.993	29.50	249.82	-1.88	yes	
2	FS	GG001	GG002	8.862	29.07	250.04	-2.05	no	farthest from the other two shots of the triple (Q8)
3	FS	GG001	GG002	9.014	29.57	249.99	-1.81	yes	
4	splay	GG001		0.595	1.95	166.96	+0.23	yes	left
5	splay	GG001		2.565	8.42	329.04	-0.04	yes	right
6	splay	GG001		0.641	2.10	344.67	+89.04	yes	up
7	splay	GG001		1.565	5.13	287.73	-84.52	yes	down
8	splay	GG001		4.785	15.70	241.15	+0.50	yes	
9	splay	GG001		7.188	23.58	278.04	-4.32	yes	
10	splay	GG001		3.670	12.04	288.18	-3.85	yes	
11	splay	GG001		1.423	4.67	310.14	-2.53	yes	
12	splay	GG001		2.762	9.06	310.10	-2.50	yes	
13	splay	GG001		3.010	9.88	83.20	-8.83	yes	
14	splay	GG001		5.046	16.56	33.40	-4.96	yes	
15	BS	GG002	GG001	9.011	29.56	69.46	+1.67	yes	backsight to GG001
16	BS	GG002	GG001	9.016	29.58	69.42	+1.61	yes	
17	splay	GG001		1.295	4.25	190.13	-19.13	no	TopoDroid azimuth quality error
18	FS	GG002	GG003	12.055	39.55	216.41	+3.07	yes	
19	FS	GG002	GG003	12.053	39.54	216.44	+3.01	yes	
20	FS	GG002	GG003	12.068	39.59	216.62	+2.95	yes	
21	splay	GG002		4.832	15.85	132.55	-7.49	yes	left
22	splay	GG002		1.527	5.01	304.48	+2.63	yes	right
23	splay	GG002		0.962	3.16	200.11	+83.28	yes	up
24	splay	GG002		0.741	2.43	208.91	-87.64	yes	down
25	splay	GG002		3.289	10.79	87.88	+6.28	yes	
26	splay	GG002		4.556	14.95	23.44	-0.01	yes	
27	splay	GG002		5.531	18.15	217.27	+3.77	yes	
28	BS	GG003	GG002	12.105	39.71	35.04	-3.11	yes	back to GG002
29	splay	GG003		11.069	36.32	91.39	-5.09	yes	was commented "back"; confirmed splay (no matching BRIC4-0038 shot)
30	splay	GG003		8.264	27.11	70.40	-5.46	yes	
31	splay	GG003		5.919	19.42	126.95	-7.13	yes	
32	splay	GG003		5.789	18.99	140.54	-6.65	yes	
33	splay	GG003		7.015	23.02	147.02	-6.33	yes	
34	splay	GG003		7.106	23.31	154.13	-5.93	yes	
35	splay	GG003		2.048	6.72	253.84	-4.85	yes	
36	splay	GG003		5.708	18.73	35.03	-2.23	yes	
37	splay	GG003		6.042	19.82	129.99	-5.98	yes	left
38	splay	GG003		1.185	3.89	330.98	+1.10	yes	right
39	splay	GG003		0.301	0.99	152.71	+77.22	yes	up
40	splay	GG003		1.065	3.49	23.41	-82.28	yes	down
41	FS	GG003	GG004	11.455	37.58	225.05	+2.09	yes	
42	FS	GG003	GG004	11.428	37.49	224.96	+1.80	yes	
43	FS	GG003	GG004	11.459	37.60	225.19	+2.12	yes	
44	FS	GG004	GG005	7.965	26.13	209.07	+21.34	yes	
45	FS	GG004	GG005	7.967	26.14	209.18	+21.30	yes	
46	FS	GG004	GG005	7.962	26.12	209.21	+21.30	yes	
47	BS	GG004	GG003	11.303	37.08	43.13	-2.12	yes	back to GG003
48	BS	GG004	GG003	11.400	37.40	42.80	+0.20	no	hand-typed copy of BRIC4-0038 shot 7 (clino typed +0.20, instrument re
49	splay	GG004		6.305	20.69	129.60	-5.80	yes	left

Shot	Role	From	To	Dist (m)	Dist (ft)	Azi (mag)	Clino	Used	Note
50	splay	GG004		1.035	3.40	303.00	-1.70	yes	right
51	splay	GG004		0.378	1.24	90.25	+83.30	yes	up
52	splay	GG004		0.748	2.45	89.06	-87.97	yes	down
53	splay	GG004		9.540	31.30	81.64	-5.83	yes	
54	splay	GG004		9.550	31.33	91.48	-5.52	yes	
55	splay	GG004		7.844	25.73	26.00	-5.01	yes	
56	splay	GG004		4.239	13.91	13.54	-8.11	yes	
57	splay	GG004		2.449	8.03	2.35	-9.39	yes	
58	splay	GG004		1.819	5.97	271.86	+24.09	yes	
59	BS	GG005	GG004	8.028	26.34	28.21	-21.76	yes	back to 4
60	FS	GG005	GG006	6.386	20.95	172.04	+12.82	yes	
61	FS	GG005	GG006	6.386	20.95	172.03	+12.82	yes	
62	FS	GG005	GG006	6.379	20.93	172.08	+12.81	yes	
63	splay	GG005		4.178	13.71	103.07	+2.55	yes	left
64	splay	GG005		4.696	15.41	292.94	+7.92	yes	right
65	splay	GG005		1.519	4.98	91.78	+85.40	yes	up
66	splay	GG005		1.476	4.84	99.24	-85.51	yes	down
67	FS	GG006	GG007	7.871	25.82	307.80	+18.62	yes	
68	FS	GG006	GG007	7.864	25.80	307.77	+18.68	yes	
69	FS	GG006	GG007	7.863	25.80	307.76	+18.57	yes	
70	splay	GG006		1.160	3.81	105.71	-2.17	yes	left
71	splay	GG006		8.581	28.15	296.67	+19.73	yes	right
72	splay	GG006		0.944	3.10	195.41	+82.40	yes	up
73	splay	GG006		0.900	2.95	28.50	-83.32	yes	down
74	BS	GG006	GG005	6.388	20.96	350.86	-12.98	yes	back
75	FS	GG006	GG008	5.141	16.87	212.21	-2.29	no	farthest from the other two shots of the triple (Q8)
76	FS	GG006	GG008	5.075	16.65	212.28	-2.54	yes	
77	FS	GG006	GG008	5.069	16.63	212.42	-2.51	yes	
78	splay	GG006		2.444	8.02	224.38	-14.75	no	cannot be clearly associated (Q7)
79	splay	GG006		6.574	21.57	41.58	-35.82	no	cannot be clearly associated (Q7)
80	splay	GG006		0.579	1.90	152.01	+82.70	no	cannot be clearly associated (Q7)
81	splay	GG006		2.289	7.51	240.03	-87.29	no	cannot be clearly associated (Q7)
82	splay	GG007		3.869	12.69	181.27	-10.66	no	cannot be clearly associated (Q7)
83	splay	GG007		6.866	22.53	40.15	-37.25	no	cannot be clearly associated (Q7)
84	BS	GG008	GG006	5.142	16.87	30.81	+2.60	yes	back to 6
85	splay	GG008		2.849	9.35	132.16	+0.38	yes	left is 2 feet
86	splay	GG008		1.202	3.94	314.80	-2.24	yes	right
87	splay	GG008		0.596	1.96	24.00	+85.16	yes	up
88	splay	GG008		0.462	1.52	326.91	-87.47	yes	down
89	FS	GG008	GG009	1.810	5.94	269.59	+18.88	yes	
90	FS	GG008	GG009	1.812	5.94	269.59	+18.47	yes	
91	FS	GG008	GG009	1.809	5.94	269.58	+18.45	yes	
92	BS	GG009	GG008	1.791	5.88	89.12	-19.55	yes	back to 8
93	FS	GG009	GG010	2.631	8.63	175.32	+50.33	yes	
94	FS	GG009	GG010	2.630	8.63	174.98	+50.25	yes	
95	FS	GG009	GG010	2.630	8.63	175.11	+50.11	yes	
96	FS	GG010	GG011	2.211	7.25	170.48	+27.01	yes	
97	FS	GG010	GG011	2.209	7.25	169.98	+26.88	yes	
98	FS	GG010	GG011	2.208	7.24	170.20	+26.90	yes	
99	splay	GG010		0.492	1.61	348.53	-26.28	yes	
100	splay	GG010		1.353	4.44	349.34	-27.17	yes	

Appendix B: raw shots, BRIC4-0038

Downloaded from the instrument's memory without station names; From/To show the leg each shot was matched to.

Shot	Role	From	To	Dist (m)	Dist (ft)	Azi (mag)	Clino	Used	Note
1	FS	GG001	GG002	9.002	29.53	249.47	-1.95	yes	1: [15]: Azimuth 1.1669233 2: [0]: No error
2	BS	GG002	GG001	9.013	29.57	70.52	+1.89	yes	1: [15]: Azimuth 1.6893045 2: [0]: No error
3	FS	GG002	GG003	12.055	39.55	214.92	+3.31	yes	
4	BS	GG003	GG002	12.040	39.50	35.61	-3.07	yes	1: [15]: Azimuth 3.1013656 2: [0]: No error
5	FS	GG003	GG004	11.401	37.40	224.48	+2.28	yes	1: [15]: Azimuth 1.6890787 2: [0]: No error
6	FS	GG004	GG005	7.981	26.18	207.14	+21.32	yes	1: [15]: Azimuth 1.7606167 2: [0]: No error
7	BS	GG004	GG003	11.408	37.43	42.81	-1.97	yes	1: [15]: Azimuth 2.473759 2: [0]: No error
8	BS	GG005	GG004	7.958	26.11	28.31	-21.65	yes	1: [15]: Azimuth 3.0171943 2: [0]: No error
9	FS	GG005	GG006	6.383	20.94	170.68	+13.24	yes	1: [15]: Azimuth 1.4549807 2: [0]: No error
10	FS	GG006	GG007	7.866	25.81	307.22	+18.85	yes	
11	?			0.317	1.04	305.81	-48.26	no	misfire (0.32 m, not a survey leg)
12	FS	GG006	GG007	7.895	25.90	307.69	+18.94	yes	1: [15]: Azimuth 1.1869084 2: [0]: No error
13	?			6.326	20.75	351.93	-16.46	no	second GG006->GG005 backsight disagrees (clino -16.5 vs -12.9); kept s
14	BS	GG006	GG005	6.378	20.93	351.04	-12.91	yes	1: [15]: Azimuth 1.0374526 2: [0]: No error
15	FS	GG006	GG008	5.147	16.89	211.03	-2.31	yes	
16	BS	GG008	GG006	5.086	16.69	30.54	+1.97	yes	
17	BS	GG008	GG006	5.157	16.92	30.54	+2.54	yes	1: [15]: Azimuth 1.9649923 2: [0]: No error
18	FS	GG008	GG009	1.831	6.01	266.49	+19.30	yes	
19	BS	GG009	GG008	1.791	5.88	89.97	-19.30	yes	1: [15]: Azimuth 1.2935182 2: [0]: No error
20	FS	GG009	GG010	2.610	8.56	174.66	+50.36	yes	
21	BS	GG010	GG009	2.618	8.59	355.36	-51.51	yes	
22	FS	GG010	GG011	2.198	7.21	170.10	+26.81	yes	
23	?			2.734	8.97	147.64	+11.22	no	no matching leg or station

Appendix C: survey and GIS files

File	Format	Opens in
Wades_GG_2026-10-31.dat	Compass survey file	Compass; CaveWhere (import)
Wades_GG_2026-10-31.svx / .3d	Survex source / processed	Survex, Aven (3D); CaveWhere (import)
Wades_GG_2026-10-31.th	Therion	Therion
legs_reduced.csv, stations.csv, shots_assigned.csv	tables	any spreadsheet
GIS/data/cave/wades_cave.gpkg	GeoPackage (old lineplot, 2026 legs, stations, splays)	QGIS, ArcGIS
Wades_Topo_Lineplot.qgz	QGIS project	QGIS

PLACEHOLDER · Formats to decide

- Walls (.srv/.wpj), CaveWhere project, KML/GeoJSON and others, to discuss.

Change log

Version	Date	Changes
0.1 draft	2026-11-02	First layout draft: data from trip 1 (2026-10-31); placeholders for history, team, trip report, sketches, photos, leads. Photo log: 12 OLDMAP photo points and photos from trip 1, photo-location map.